

Engineering Mathematics III - Lecture Notes

Course Code: MATH301 | Department of Computer Science & Engineering

1. Linear Algebra: Matrix Decompositions & Eigenvalues

A symmetric matrix A in $\mathbb{R}^{(n \times n)}$ can be orthogonally diagonalized as $A = Q * \Lambda * Q^T$.

Here, Q is an orthogonal matrix whose columns are orthonormal eigenvectors of A ,

and Λ is a diagonal matrix containing the real eigenvalues of A .

Key property: $\text{Trace}(A)$ equals the sum of eigenvalues, and $\text{Det}(A)$ equals their product.

2. Vector Calculus & Field Theorems

Gradient: $\text{grad}(f) = [df/dx, df/dy, df/dz]$ points in the direction of greatest rate of increase.

Divergence Theorem: $\text{Integral over boundary } dV \text{ of } (F \cdot n \, dS) = \text{Triple integral over } V \text{ of } \text{div}(F) \, dV$.

Stokes Theorem: $\text{Closed loop line integral of } (F \cdot dr) = \text{Surface integral of } (\text{curl}(F) \cdot n \, dS)$.

Conservative Fields: A field F is conservative if $\text{curl}(F) = 0$ and line integral is path independent.

3. Numerical Methods & Root Finding

Newton-Raphson formula: $x_{(n+1)} = x_n - f(x_n) / f'(x_n)$ exhibits quadratic convergence.

Runge-Kutta 4th Order (RK4): Widely utilized for solving first-order ODEs with high precision.